Last Update: August 5, 2026

# Yeonsu Kwak

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## EDUCATION

***Ph.D. Candidate*** in Chemical & Biomolecular Engineering — Aug 2021 – Aug 2026 (Expected)
University of Delaware (UD), Newark, DE, United States
***M.S.*** in Energy — Sep 2016 – Aug 2017
Texas A&M University (TAMU), College Station, TX, United States
***B.S.*** in Chemical Engineering — Mar 2012 – Aug 2016
Pohang University of Science and Technology (POSTECH), Pohang, South Korea

## RESEARCH INTERESTS

Heterogeneous catalysis | Catalytic reaction engineering | Biomass, plastic, and hydrocarbon conversion
Continuous-flow reactors | Catalyst characterization and operando spectroscopy | Process intensification & electrification

## RESEARCH EXPERIENCE

- **Catalyst & Reactor Engineering:** Catalyst synthesis, redox processing, high-pressure flow systems, thermal treatment, continuous/batch reactors, operando and in situ experimentation
- **Process Development:** Lab-to-pilot scale-up, high-temperature systems, electrified reactor design (microwave/Joule), thermal management, continuous processing, technoecnomic analysis, thermodynamic/kinetic calculations
- **Process Modeling & Analysis:** UniSim, Aspen Plus, Pro/II, ProMax, Python, Autodesk Fusion
- **Characterization:** XPS, AP-XPS, XAS (trained in BNL), XRD, TEM/STEM, SEM, BET, GC-FID/TCD, GC-MS, QMS, Raman, UV-Vis, FT-IR/DRIFTS, TGA/TPD/TPR, dual-mode dielectric property measurement

## RESEARCH EXPERIENCE

**R&D Intern** — May 2026 – Present
Linde Engineering America, Tonawanda, NY
- Developed and validated an equilibrium-kinetic model and interactive tool that use reactor operating conditions to predict high-temperature water-gas shift performance and hydrocarbon byproduct formation rates
- Developed a UniSim model of a hydrogen plant flowsheet and established quantitative design guidance for $H_2$ recycle, reformer and shift configurations, heat integration, and carbon intensity

**Graduate Research Assistant** — Aug 2021 – Aug 2026 (Expected)
University of Delaware, Newark, DE, United States (Advisor: Prof. Dionisios G. Vlachos)
- Designed heterogeneous catalysts and investigated structure–property–performance relationships using operando dielectric, Raman, XRD, and catalytic measurements to understand oxygen-vacancy-mediated reactivity
- Developed continuous-flow microwave and conventional catalytic reactor systems for methane reforming, propane dehydrogenation, and ethane cracking
- Combined pulsed Joule-heating experiments with microkinetic modeling conducted in collaboration to explain enhanced methane coupling and reduced coke and aromatic byproduct formation relative to continuous heating
- Led experimental reactor development and collaborated with CFD researchers to design an internally Joule-heated ethane cracker with enhanced ethylene yield and projected reductions in capital requirements and $CO_2$ emissions
- Synthesized $PtSn/SiO_2$ catalysts and demonstrated coke-resistant microwave propane dehydrogenation (PDH) without $H_2$ co-feed, with higher energy efficiency and slower nanoparticle sintering than furnace heating

**Visiting Scholar** — Jun 2024
ITACA Institute, Universitat Politècnica de València, Spain (Host: Prof. José M. Catalá-Civera)
- Developed perturbation-based dual-mode cavity protocols to quantify microwave heating profiles and dielectric properties for operando catalysis studies
- Coordinated ITACA-UD dual mode cavity co-development, international shipment and customs, and preparation for XAS-compatible microwave experiments at QAS, BNL

**Visiting Scholar** — Nov 2022
INMA Institute, University of Zaragoza, Spain (Host: Prof. Jesús Santamaria)

- Conducted cross-laboratory experiments to verify the reproducibility of microwave-assisted PDH performance

**Research Scientist (Alternative Military Service)** Nov 2019 – Aug 2021
Korea Institute of Science and Technology (KIST), Seoul, South Korea

- Advanced LOHC dehydrogenation systems from catalyst benchmarking to low-platinum-loading catalyst development via atomic layer deposition for scalable hydrogen release
- Designed and constructed Ni-alloy foam architectures for Joule-heated ammonia decomposition reactors
- Scaled an LOHC dehydrogenation reactor to 2.3 $Nm^3\ hr^{-1}$ $H_2$ production using phase-change thermal management
- Managed multi-year projects funded by Hyundai Motor Company and National Research Foundation of Korea

**Research Engineer (Alternative Military Service)** Aug 2017 – Nov 2019
Institute for Advanced Engineering (IAE), Yongin, South Korea

- Scaled a continuous 5 ton $yr^{-1}$ sustainable aviation fuel pilot process integrating palm-oil deoxygenation, hydrocracking, isomerization, and distillation to produce jet-fuel-range hydrocarbons meeting ASTM D7566
- Managed continuous pilot operations, rotating shift schedules, and process-safety procedures

**Graduate Research Assistant** Sep 2016 – Aug 2017
Texas A&M University, College Station, TX, United States (Advisors: Profs. Hae-Kwon Jeong & Phanourios Tamamis)

- Synthesized $CO_2$-capturing amyloid biomaterials via peptide self-assembly and validated $CO_2$ uptake

## PUBLICATIONS

(†Equal contribution, *Corresponding authors, Full list of publications)

**First-author Publications**

[10] **Y. Kwak**†, R. Railkar†, W. Zheng & D. G. Vlachos*
"Tandem non-oxidative coupling of methane and carbon dioxide reduction via pulsed Joule thermochemistry"
*ACS Energy Letters,* 2025, 10, 12, 6188–6196.

[9] A. Mittal†, **Y. Kwak**†, W. Zheng, M. Ierapetritou & D. G. Vlachos*
"Short contact time, high-temperature internally-heated ethane cracker"
*Chemical Engineering Journal* 523 (2025) 168251.

[8] **Y. Kwak**†, Y.-J. Lee†, S. Moon†, K. Lee, S. Ramadhani, E.-R. On, C.-I. Ahn, S.-J Hwang, H. Sohn, H. Jeong, S. W. Nam & Y. Kim*
"Scalable atomic-layer tailoring of abundant oxide supports unlocks superior interfaces for low-metal-loading dehydrogenation"
*Angewandte Chemie International Edition* 2025, 64, e202417598.

[7] **Y. Kwak**, C. Wang, K. C. Ajit, K. Yu, E. Selvam, R. Mallada, J. Santamaria, H. Goyal, I. Julian, J. M. Catala-Civera, W. Zheng & D. G. Vlachos*
"Microwave-assisted, performance-advantaged electrification of propane dehydrogenation"
*Science Advances*, 9 (37), eadi8219 (2023).

[6] D. K. Kang†, **Y. Kwak**†, S. Moon, S. Ramadhani, W. Jeong, H. Jeong, H. Sohn, S. W. Nam, Y. S. Jo, C. W. Yoon & Y. Kim*
"Thermally manageable and scalable reactor systems for highly efficient hydrogen release from LOHC"
*Energy Conversion and Management* 307 (2024) 118345.

[5] **Y. Kwak**, S. Moon, C.-I. Ahn, A. Kim, Y. Park, Y. Kim, H. Jeong, H. Sohn, S. W. Nam, C. W. Yoon* & Y. S. Jo*
"Effect of support properties in dehydrogenation of biphenyl-based eutectic mixture as LOHC over $Pt/Al_2O_3$ catalysts"
*Fuel* 284 (2021) 119285.

[4] **Y. Kwak**, J. Kirk, S. Moon, T. Ohm, Y.-J. Lee, M. Jang, L.-H Park, C.-I Ahn, H. Jeong, H. Sohn, S. W. Nam, C. W. Yoon, Y. S. Jo* & Y. Kim*
"Hydrogen production from homocyclic LOHCs: Benchmarking studies and energy-economic analyses"
*Energy Conversion and Management* 239 (2021) 114124.

[3] **Y. Kwak**, H. Shin, S. Moon, J. Kirk, Y. Park, H. Jeong, H. Sohn, S. W. Nam, C. W. Yoon, Y. S. Jo* & Y. Kim*
"Investigation of a hydrogen generator with the heat management module utilizing liquid-gas organic phase change material"
*International Journal of Energy Research* (2021) 45, 10378–10392.

[2] Y. Park†, **Y. Kwak**†, S. Yu, A. Badakhsh, Y.-J. Lee, H. Jeong, Y. Kim, H. Sohn, S. W. Nam, C. W. Yoon, J. H. Han* & Y. S. Jo*
"Degradation mechanism of a Pd/Ta composite membrane: Catalytic surface fouling with inter-diffusion"

*Journal of Alloys and Compounds* 854 (2021) 157196.

[1] **Y. Kwak**, J. H. Jang, S. Kim, M. Ahn, E.-S Lee, G. B. Han*, B. H. Jeong, J. S. Han & C.-H Jeon*
"Reaction characteristics of hydrocarbon fuels under various operation conditions of hydro-upgrading process for vegetable oil-based bio-jet fuel production"
*Journal of the Korean Oil Chemists Society*, 35 (2018) 731–743.

**Selected Co-author Publications (10 out of 16)**

[10] J. Choi, M. Kim, **Y. Kwak**, A. Pophali, G. Halada, H. Luo, G. Kwon, I. Ro, J. Kim, M. Rafailovich & T. Kim*
"Elucidating the role of surface species in CO oxidation catalyzed by boron nitride nanotube supported transition metal oxides"
*Journal of Catalysis* (2026) 116715.

[9] R. Railkar, **Y. Kwak** & D. G. Vlachos*
"Intensifying steam methane reforming and water-gas shift in tandem via rapid pulsed Joule heating"
*Chemical Engineering Journal* 512 (2025) 162700.

[8] E. Selvam, J. A. Sun, P. Kots, Z. Schyns, **Y. Kwak,** L. Korley, R. Lobo & D. G. Vlachos*
"Conversion of compositionally diverse plastic waste over earth-abundant sulfides"
*Journal of the American Chemical Society* 2025, 147 (13) 11227–11238.

[7] E.-R. On, K. Lee, **Y. Kwak,** C. Kim, Q. Dao, H. Sohn, S. W. Nam, J. Kim, Y. Kim* & H. Jeong*
"Promoter-guided reaction intermediate dynamics enhance hydrogenated benzyltoluene dehydrogenation"
*ACS Catalysis*, 2025, 15, 5531–5545.

[6] S. Gupta, **Y. Kwak,** R. P. Raj & P. Selvam*
"Ytterbium-nitrogen co-doped ordered mesoporous $TiO_2$: The innovative hetero-phase photocatalyst for harnessing solar energy in green hydrogen production"
*Journal of Materials Chemistry A* (2024) 12, 6906–6927.

[5] C. Wang, S. Sourav, K. Yu, **Y. Kwak,** W. Zheng & D. G. Vlachos*
"Green syngas production by microwave-assisted dry reforming of methane on doped ceria catalysts"
*ACS Sustainable Chemistry & Engineering* 2023, 11 (36) 13353–13362.

[4] I. S. Lim, Y. Jeong, **Y. Kwak,** E.-R. On, H. Jeong, H. Sohn, S. W. Nam, T.-H. Lim, K. Muller & Y. Kim*
"Maximizing clean hydrogen release from hydrogenated benzyltoluene: Energy-efficient scale-up strategies and techno-economic analyses"
*Chemical Engineering Journal* 478 (2023) 147296.

[3] C.-I Ahn, **Y. Kwak,** A. Kim, M. Jang, A. Badakhsh, Y. S. Jo, Y. Kim, H. Jeong, S. W. Nam, C. W. Yoon* & H. Sohn*
"Dehydrogenation of homocyclic liquid organic hydrogen carriers over Pt supported on an ordered pore structure of 3-D cubic mesoporous KIT-6 silica"
*Applied Catalysis B: Environment and Energy* 307 (2022) 121169.

[2] A. Badakhsh, **Y. Kwak,** Y.-J. Lee, H. Jeong, Y. Kim, S. W. Nam, C. W. Yoon, C. W. Park* & Y. S. Jo*
"A compact catalytic foam reactor for decomposition of ammonia by the Joule-heating mechanism"
*Chemical Engineering Journal* 426 (2021) 130802.

[1] Y.-J Lee, J. Cha, **Y. Kwak,** Y. Park, Y. S. Jo, H. Jeong, H. Sohn, C. W. Yoon, Y. Kim*, K.-B Kim* & S. W. Nam*
"Top-down syntheses of nickel-based structured catalyst for hydrogen production from ammonia"
*ACS Applied Materials & Interfaces* 2021, 13, 1, 597–607.

MANUSCRIPTS UNDER REVIEW, IN REVISION, AND IN PREPARATION

[3] F. Rasteiro et al., Challenges and opportunities of catalytic process with alternative forms of energy input, *Chemical Reviews, under review*

[2] **Y. Kwak** et al., Dielectric-redox coupling in reducible metal oxides for microwave-driven redox catalysis, *in preparation*

[1] **Y. Kwak** et al., From electromagnetic field-material interactions to intensified chemical manufacturing via microwave catalysis, *Chemical Society Reviews*, *under review*

## ISSUED PATENTS

- Hydrogen extraction reactor and hydrogen extraction process using phase change materials, **US 11,674,068 B2**
- A catalyst for dehydrogenation of LOHCs and method for producing the same, **KR 10-2472412**
- Apparatus for manufacturing eco-friendly transportation fuel and method therefor, **KR 10-2859154**
- Apparatus for separating and producing hydrocarbon fuels, **KR 10-2786950**

## SELECTED ORAL PRESENTATIONS

- Process Electrification for Scalable and Intensified Chemical Manufacturing, KAIST YGIS, Korea 2026
- Dynamic Electrification of Nonoxidative Propane Dehydrogenation, NAM29, GA 2025
- Electrified Biogas Upgrading into Value-Added Chemicals, AIChE Fall, CA 2024
- Mechanistic Insights into Microwave-Assisted Thermal Catalysis, 5GCMEA, Japan 2024
- Microwave-Assisted Direct Propane Dehydrogenation, AIChE Fall, FL & NAM28, PI 2023
- Electromagnetic field driven catalysis for electrification, ACS Spring, Hybrid 2023

## AWARDS & HONORS

Theodore A. Koch Graduate Student Travel Award | Catalysis Club of Philadelphia (CCP) 2026
NSF EPSCoR Graduate Bridge Fund Award 2025 – 2026
Best Oral Presentation Award | 5th Global Congress on Microwave Energy Applications (5GCMEA) 2024
AIChE Hanwha Travel Award | Hanwha Solutions & TotalEnergies Petrochemical 2024
Graduate Student Travel Awards | UD Graduate College & UD CCST 2024 – 2025
Commendation for Oral PhD Qualifier Exam (*Exceptional*) | UD ChBE 2022
Finalist of Fulbright Graduate Scholarship | US Department of State & Fulbright Korea 2021
Master of Science in Energy Scholarship | Texas A&M Energy Institute 2017
Global Engineers Fostering Scholarship (*M.S. Full Funding*) | EDRC of Seoul National University 2016 – 2017
National Scholarship in Science and Technology (*B.S. Full Funding*) | Ministry of Education of Korea 2012 – 2016
4th Place Award, Korea Chemical Process Simulation Olympiad | KIChE, Schneider Electric 2016
Research Scholarship for Undergraduate Research Program | POSTECH 2015
Exchange Student Scholarship (National University of Singapore '15 Fall) | POSTECH 2015
Exchange Student Scholarship (Nanyang Technological University '14 Summer) | POSTECH 2014

## SERVICE & LEADERSHIP

**Peer Reviewer** | American Chemical Society, Royal Society of Chemistry & Elsevier 2022 – Present

- Completed 40 peer reviews for Chemical Engineering Journal, Fuel, International Journal of Hydrogen Energy, Catalysis Today, Molecular Catalysis, ACS Omega, Journal of Alloys and Compounds, RSC Advances, and Fuel Processing Technology

**Research Mentor** | UD Delaware Energy Institute 2023 – Present

- Auden Jones (‘24-‘25): Microwave catalysis of metal oxides; admitted to U of Houston's Ph.D. program
- Matthew Conlon (‘23): Microwave heating of carbon materials; admitted to Purdue's Ph.D. program

**President (‘23-‘24), Treasurer (‘22-‘23)** | UD Korean Graduate Student Association 2022 – 2024

- Managed annual finances exceeding $3,000 and organized 10+ academic and community events, serving 100+ members of the UD Korean and Korean-American community

**Teaching Assistant** | UD ChBE 2023

- CHEG614/814 (Special Topics in Energy, Spring), Instructor: Prof. Dionisios G. Vlachos. Led office hours and facilitated graduate-level discussions on energy systems
- CHEG445 (Chemical Engineering Lab II - Distillation, Fall), Instructor: Prof. Nicholas Shevchenko. Supervised distillation pilot plant operations and graded technical reports for 30+ seniors, adhering to safety protocols

## PROFESSIONAL MEMBERSHIP

Catalysis Club of Philadelphia (CCP), Graduate Student Member 2024 – Present
American Institute of Chemical Engineers (AIChE), Graduate Student Member 2022 – Present
Korean-American Scientists and Engineers Association (KSEA), Graduate Student Member 2022 – Present
Korean Institute of Chemical Engineers (KIChE), Regular Member (On Leave) 2017 – Present

## REFERENCES WITH CONTACT INFORMATION

Dionisios G. Vlachos, Ph.D. | Ph.D. Advisor | Professor, University of Delaware, USA | vlachos@udel.edu
Raul Lobo, Ph.D. | Ph.D. Dissertation Committee | Professor, University of Delaware, USA | lobo@udel.edu
José M. Catalá-Civera, Ph.D. | Research Collaborator | Professor, ITACA Institute, Spain | jmcatala@dcom.upv.es
Yongmin Kim, Ph.D. | Direct Supervisor at KIST | Principal Researcher, KIST, South Korea | yongminkim@kist.re.kr